

Program – 9: Organizing Kubernetes Pods Using Namespaces and Labels

Aim:

To create and manage Kubernetes pods using namespaces and labels in a Minikube cluster on Ubuntu.

Requirements:

- Ubuntu system
- Minikube
- kubectl
- namespaces.yaml, pods.yaml

Step 1: Start Minikube

```
minikube start
```

Step 2: Create Namespaces

```
namespaces.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
kubectl apply -f namespaces.yaml
kubectl get namespaces
```

Step 3: Create Pods with Labels

```
pods.yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-podnamespace: dev
labels:
  app: frontend
  env: dev
spec:
  containers:
  - name: frontend
    image: nginx
  ports:
  - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
labels:
```

```
  app: backend
env: dev
spec:
containers:
- name: backend
image: nginx
ports:
- containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
name: frontend-pod
namespace: prodlabels:
app: frontend
env: prod
spec:
containers:
- name: frontend
image: nginx
ports:
- containerPort: 80
apiVersion: v1
kind: Pod
metadata:
name: backend-pod
namespace: prod
labels:
app: backend
env: prod
spec:
containers:
- name: backend
image: nginx
ports:
- containerPort: 80
```

kubectyl apply -f pods.yaml

step4:filter pods using namespace

kubectyl get pods -n dev

kubectyl get pods -n prod

Step 5: Filter Pods Using Labels

kubectyl get pods -n dev -l app=frontend

kubectyl get pods -n prod -l app=backend

```
1rv24mc074_poornimat@lenovo: ~/program9
1rv24mc074_poornimat@lenovo:~/program9$ ls
namespaces.yaml  pods.yaml
1rv24mc074_poornimat@lenovo:~/program9$ kubectl apply -f namespaces.yaml
namespace/dev unchanged
namespace/prod unchanged
1rv24mc074_poornimat@lenovo:~/program9$ kubectl apply -f pods.yaml
pod/frontend-pod unchanged
pod/backend-pod unchanged
pod/frontend-pod unchanged
pod/backend-pod unchanged
1rv24mc074_poornimat@lenovo:~/program9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   4 (18m ago) 19d
backend-pod1   1/1     Running   4 (18m ago) 19d
frontend-pod   1/1     Running   4 (18m ago) 19d
frontend-pod1  1/1     Running   4 (18m ago) 19d
1rv24mc074_poornimat@lenovo:~/program9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   4 (19m ago) 19d
backend-pod1   1/1     Running   4 (19m ago) 19d
frontend-pod   1/1     Running   4 (19m ago) 19d
frontend-pod1  1/1     Running   4 (19m ago) 19d
1rv24mc074_poornimat@lenovo:~/program9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   4 (19m ago) 19d
1rv24mc074_poornimat@lenovo:~/program9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   4 (19m ago) 19d
1rv24mc074_poornimat@lenovo:~/program9$
```